

## Deep Learning Ex 2: Iris Prediction

Code:

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import matplotlib.pyplot as plt
import numpy as np
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense

iris = load_iris()
X = iris.data
y = iris.target.reshape(-1, 1)
classes = iris.target_names

encoder = OneHotEncoder(sparse_output=False)
y_encoded = encoder.fit_transform(y)

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(X_scaled,
y_encoded, test_size=0.2, random_state=42)

def create_mlp(input_dim, output_dim, hidden_layers,
activation='relu'):
    model = Sequential()
    model.add(Dense(hidden_layers[0], input_dim=input_dim,
activation=activation))
    for units in hidden_layers[1:]:
        model.add(Dense(units, activation=activation))
    model.add(Dense(output_dim, activation='softmax'))
    model.compile(optimizer='adam', loss='categorical_crossentropy',
metrics=['accuracy'])
    return model

hidden_layer_configs = [[8], [16, 8], [32, 16, 8]]
activations = ['relu', 'tanh', 'sigmoid']

for hidden_layers in hidden_layer_configs:
    for activation in activations:
        print(f"\nTesting MLP with hidden_layers={hidden_layers},
activation={activation}")
        model = create_mlp(input_dim=4, output_dim=3,
hidden_layers=hidden_layers, activation=activation)
        history = model.fit(X_train, y_train, epochs=50, batch_size=5,
verbose=0, validation_split=0.1)
```

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        test_loss, test_acc = model.evaluate(X_test, y_test, verbose=0)
        print(f"Test Accuracy: {test_acc:.4f}, Test Loss:
{test_loss:.4f}")

        plt.figure(figsize=(10, 4))
        plt.suptitle(f"Config: {hidden_layers}, Activation:
{activation}", fontsize=14)

        plt.subplot(1, 2, 1)
        plt.plot(history.history['accuracy'], label='Train Acc')
        plt.plot(history.history['val_accuracy'], label='Val Acc')
        plt.xlabel('Epoch')
        plt.ylabel('Accuracy')
        plt.title('Model Accuracy')
        plt.legend()

        plt.subplot(1, 2, 2)
        plt.plot(history.history['loss'], label='Train Loss')
        plt.plot(history.history['val_loss'], label='Val Loss')
        plt.xlabel('Epoch')
        plt.ylabel('Loss')
        plt.title('Model Loss')
        plt.legend()

        plt.tight_layout()
        plt.show()

sepal_length = float(input("Sepal length (cm): "))
sepal_width = float(input("Sepal width (cm): "))
petal_length = float(input("Petal length (cm): "))
petal_width = float(input("Petal width (cm): "))

user_input = np.array([[sepal_length, sepal_width, petal_length,
petal_width]])
user_input_scaled = scaler.transform(user_input)

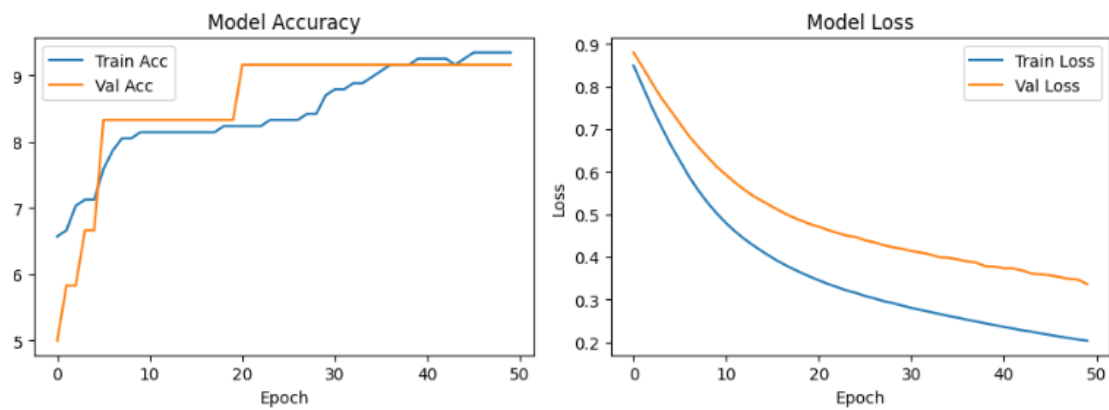
prediction = model.predict(user_input_scaled)
predicted_class_index = np.argmax(prediction)
predicted_class_name = classes[predicted_class_index]

print(f"\n Predicted Iris Species: {predicted_class_name}")

```

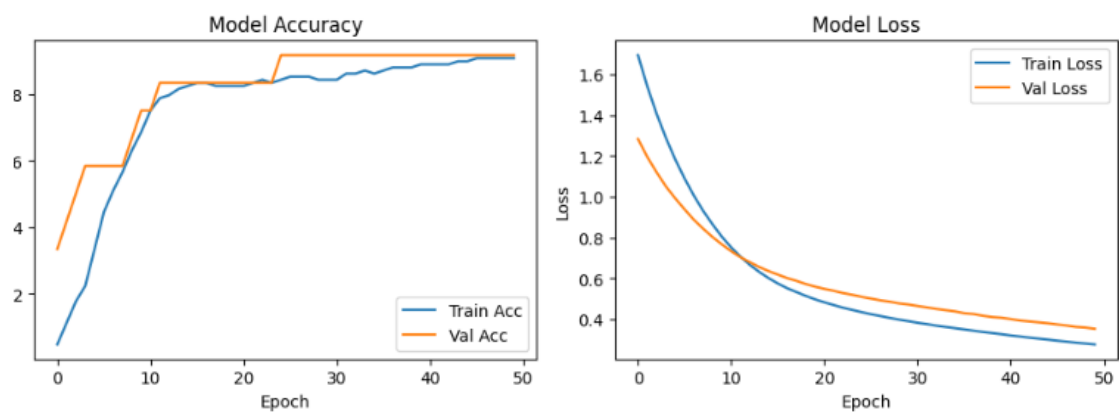
Output:

Config: [8], Activation: relu



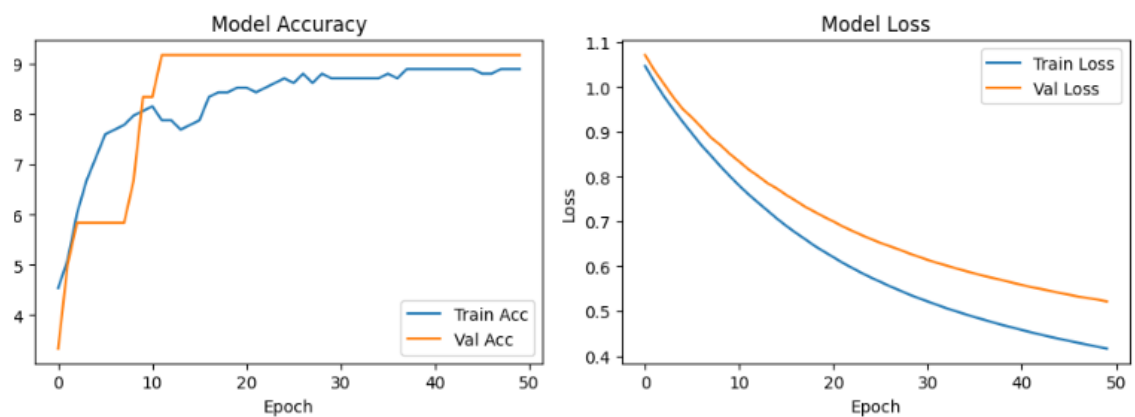
ig MLP with hidden\_layers=[8], activation=tanh  
accuracy: 0.9667, Test Loss: 0.2217

Config: [8], Activation: tanh



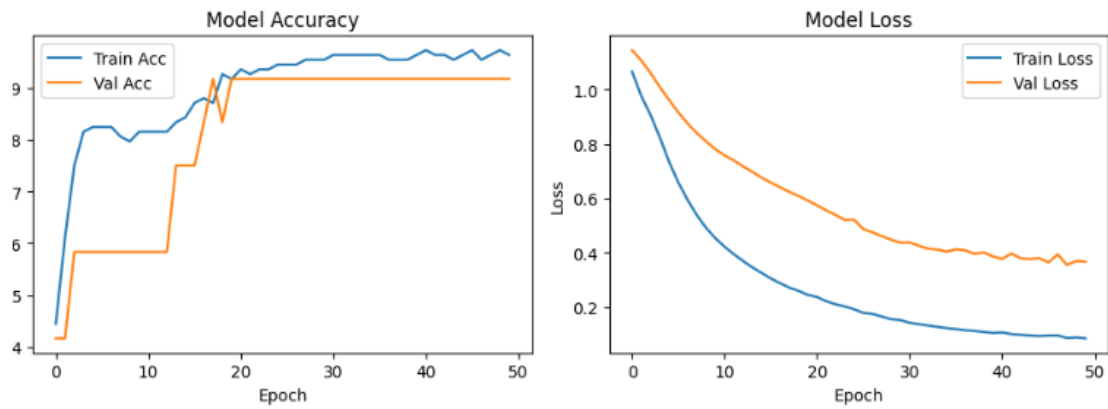
ig MLP with hidden\_layers=[8], activation=sigmoid  
accuracy: 0.9667, Test Loss: 0.3948

Config: [8], Activation: sigmoid



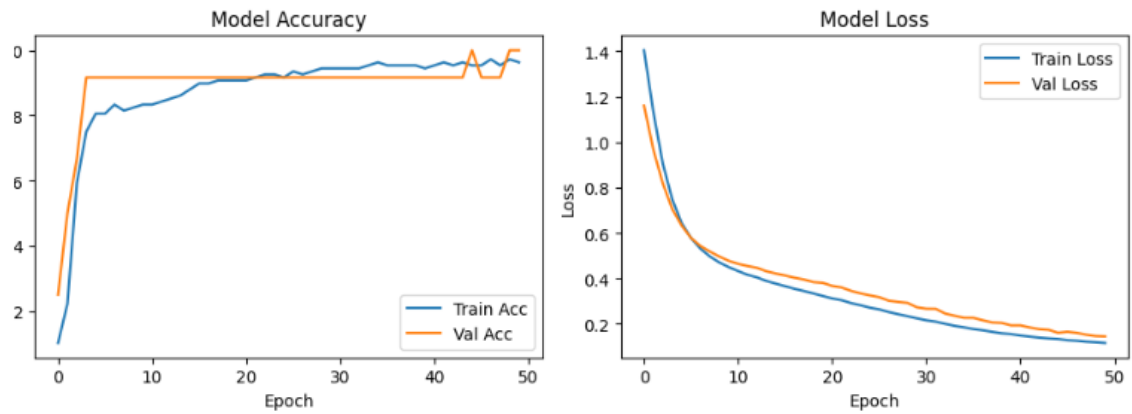
ig MLP with hidden\_layers=[16, 8], activation=relu  
accuracy: 0.9667, Test Loss: 0.0856

Config: [16, 8], Activation: relu



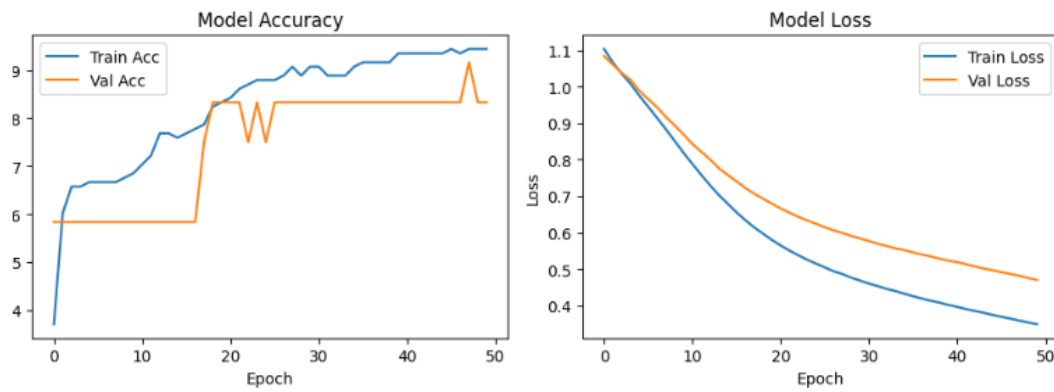
ig MLP with hidden\_layers=[16, 8], activation=tanh  
accuracy: 1.0000, Test Loss: 0.0943

Config: [16, 8], Activation: tanh



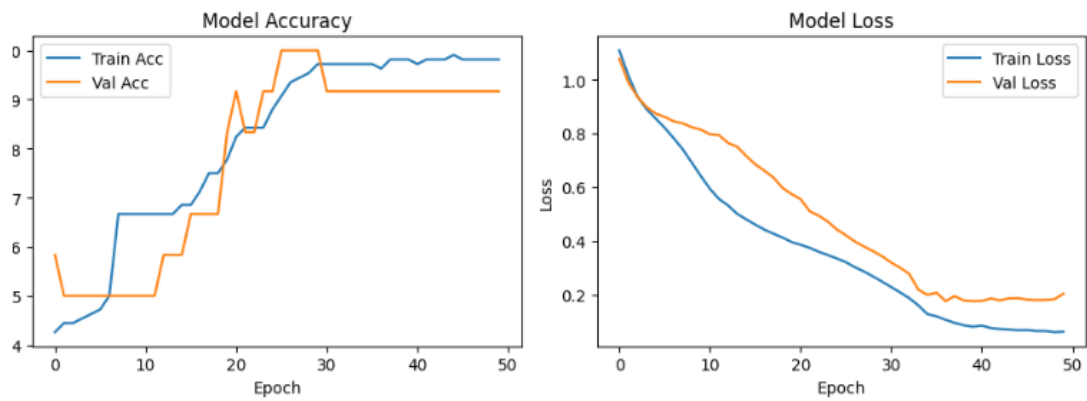
ig MLP with hidden\_layers=[16, 8], activation=sigmoid  
accuracy: 0.9333, Test Loss: 0.3435

Config: [16, 8], Activation: sigmoid



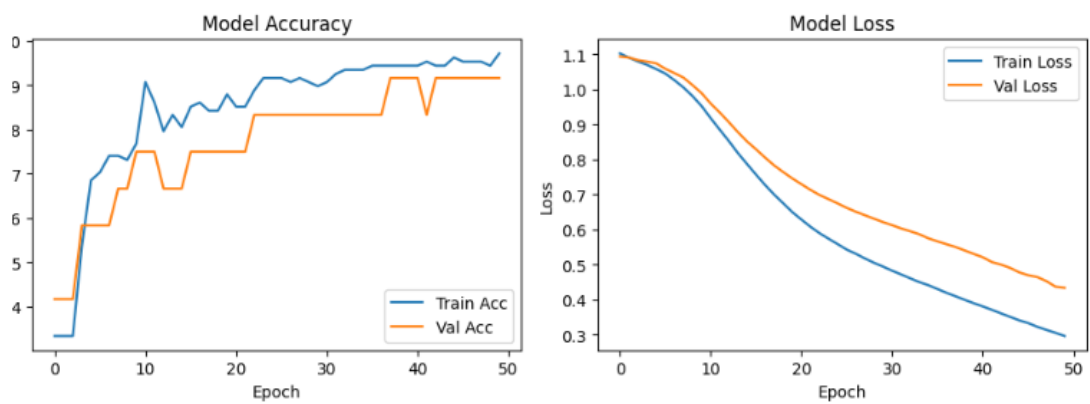
ig MLP with hidden\_layers=[32, 16, 8], activation=relu  
accuracy: 1.0000, Test Loss: 0.0309

Config: [32, 16, 8], Activation: relu



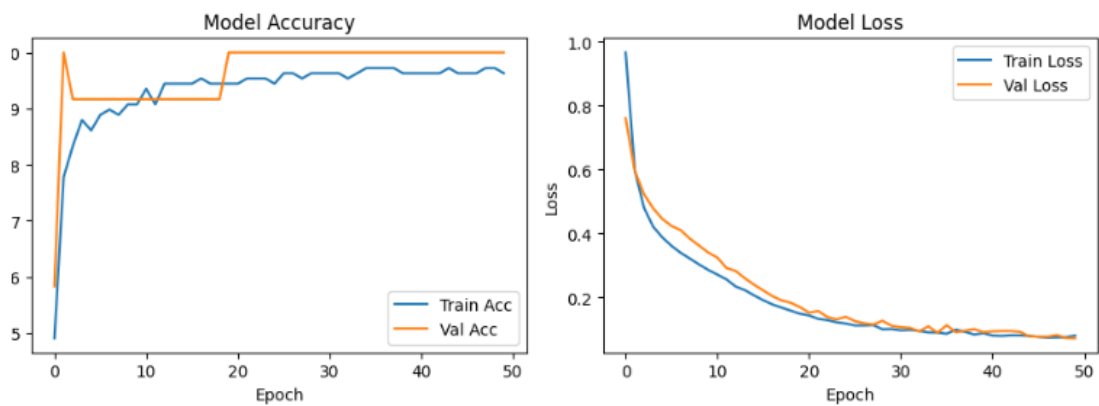
ig MLP with hidden\_layers=[32, 16, 8], activation=tanh  
accuracy: 1.0000, Test Loss: 0.0498

Config: [32, 16, 8], Activation: sigmoid



length (cm):

Config: [32, 16, 8], Activation: tanh



ig MLP with hidden\_layers=[32, 16, 8], activation=sigmoid  
accuracy: 0.9333, Test Loss: 0.2974